MCS Saturday 7th-SHSAT

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Name _____ Date _____

Lines & Linear Models

Session 4 · Digital SAT Math

Homework · From structure to strategy

A · Lines	B · Models & functions	C · Coordinates	D · Mixed sprint
6 questions · 12 min	6 questions · 14 min	4 questions · 9 min	8 questions · 13 min

Complete Parts A–C thoughtfully; use a calculator when useful and show a setup or explanation. Then complete Part D without notes. Allow about 60–65 minutes including review. This is targeted practice, not a full adaptive test. L1–L4 are instructor difficulty labels.

A Decode and build lines

1–4 of 6

Question 1 L2 · MC

What is the slope of the line represented by **9x + 6y = 24**?

- (●) **A** −3/2
- ( ) **B** −2/3
- ( ) **C** 3/2
- ( ) **D** 4

Show your reasoning or calculator setup.

6y = -9x + 24
y = (-3/2)x + 4; slope = -3/2.

☐ Flag for review

Question 2 L2 · MC

The table gives three points on a line. Which equation represents the line?

Values on a line

x	y
−4	11
0	3
6	−9

- ( ) **A** y = 2x + 3
- ( ) **B** y = −2x + 11
- (●) **C** y = −2x + 3
- ( ) **D** y = 3x − 2

Show your reasoning or calculator setup.

m = (-9 - 3)/(6 - 0) = -2.
At x = 0, y = 3, so y = -2x + 3.

☐ Flag for review

Question 3 L2 · MC

Which equation represents a line parallel to the line **5x − 2y = 8**?

- ( ) **A** y = (−2/5)x + 6
- (●) **B** y = (5/2)x − 7
- ( ) **C** y = (−5/2)x − 7
- ( ) **D** y = (2/5)x + 6

Show your reasoning or calculator setup.

5x - 2y = 8 gives y = (5/2)x - 4.
Parallel lines have the same slope, 5/2.

☐ Flag for review

Question 4 L3 · MC

A line is perpendicular to **3x + 4y = 20** and passes through (6, −1). Which equation represents this line?

- ( ) **A** y = (−4/3)x + 7
- ( ) **B** y = (3/4)x − 11/2
- ( ) **C** y = (4/3)x + 7
- (●) **D** y = (4/3)x − 9

Show your reasoning or calculator setup.

Original slope = -3/4; perpendicular = 4/3.
y + 1 = (4/3)(x - 6)
y = (4/3)x - 9.

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A Lines, continued

5–6 of 6

Student-produced responses (SPR): Enter only the requested number—no units, variables, or ordered pairs. Use an improper fraction rather than a mixed number. Positive entries may use up to 5 characters; negative entries may use up to 6 including the minus sign. Exact fractions work well for this sheet.

Question 5 L4 · SPR

The line **ax + 8y = 24** is perpendicular to the line **y = 2x − 5**. What is the value of a?

Answer **4**

One number; use a fraction or decimal if needed.

Show your reasoning or calculator setup.

The slope of ax + 8y = 24 is -a/8.
(-a/8)(2) = -1, so a = 4.

☐ Flag for review

Question 6 L2 · MC

A line passes through (3, 8) and is perpendicular to the line **y = −6**. Which equation represents the line?

- ○ **A** y = 3
- ◉ **B** x = 3
- ○ **C** y = 8
- ○ **D** x = 8

Show your reasoning or calculator setup.

y = -6 is horizontal; its perpendicular is vertical. Through (3, 8), the line is x = 3.

☐ Flag for review

B Interpret linear models

7–8 of 12

For each model, distinguish the starting amount, the rate, and the requested change or total.

Question 7 L2 · MC

An equipment rental company models the cost, in dollars, of renting equipment for h hours by **C(h) = 27 + 14h**. What does 27 represent?

- ○ **A** The rental cost per hour
- ○ **B** The number of hours included in the rental
- ◉ **C** A fixed fee independent of the rental duration
- ○ **D** The cost of a 27-hour rental

Show your reasoning or calculator setup.

C(0) = 27, so $27 is charged even at 0 hours.
It is the fixed fee, not the hourly rate.

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Question 8 L2 · MC

For $0 \le t \le 20$, a cooling process is modeled by **T(t) = 62 − 1.4t**, where T is temperature in degrees Celsius and t is time in minutes. By how many degrees Celsius does the temperature decrease from t = 3 to t = 8?

- ○ **A** 1.4
- ◉ **B** 7
- ○ **C** 50.8
- ○ **D** 57.8

Show your reasoning or calculator setup.

Elapsed time = 8 - 3 = 5 minutes.
Temperature decrease = 1.4(5) = 7 degrees C.

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B Models and function inputs

9–12 of 24

Keep the entire input together when substituting. A request for an input calls for an equation, not forward evaluation. Question 11 extends the lesson’s inverse-function notation.

Question 9

L3 · SPR

A printing service charges a fixed setup fee plus a constant price per poster. An order of 4 posters costs $86, and an order of 9 posters costs $146. What is the setup fee, in dollars?

Answer: **38**

One number; use a fraction or decimal if needed.

Show your reasoning or calculator setup.

Price per poster = (146 - 86)/(9 - 4) = 12.
Setup fee = 86 - 12(4) = 38.

☐ Flag for review

Question 10

L3 · MC

The function f is defined by **f(x) = 7 − 3x**. Which expression is equivalent to f(2a − 4)?

- (●) **A** 19 − 6a
- ( ) **B** 3 − 6a
- ( ) **C** −5 − 6a
- ( ) **D** 14a − 31

Show your reasoning or calculator setup.

f(2a - 4) = 7 - 3(2a - 4)
= 7 - 6a + 12 = 19 - 6a.

☐ Flag for review

Question 11

L3 · SPR

The function g is defined by **g(x) = (3x + 5)/4**. What is the value of $g^{-1}(8)$?

Answer: **9**

One number; use a fraction or decimal if needed.

Show your reasoning or calculator setup.

Solve g(x) = 8: (3x + 5)/4 = 8.
3x + 5 = 32; 3x = 27; x = 9.

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Question 12

L3 · MC

For a linear function f, **f(2) = 11** and **f(8) = 29**. What is the value of f(x + 1) − f(x) for any real number x?

- ( ) **A** 18
- ( ) **B** 6
- ( ) **C** 5
- (●) **D** 3

Show your reasoning or calculator setup.

m = (29 - 11)/(8 - 2) = 3.
A 1-unit input increase raises the output by 3.

☐ Flag for review

C Coordinate geometry connections

4 questions · 9 min

Match x with x and y with y. On a separate sketch, label the point or length requested before calculating. A perpendicular bisector passes through a segment’s midpoint at a right angle.

Question 13

L2 · MC

What is the midpoint of the segment with endpoints (−7, −4) and (5, 10)?

- ○ **A** (−2, 6)
- ◉ **B** (−1, 3)
- ○ **C** (1, 7)
- ○ **D** (6, 7)

Show your reasoning or calculator setup.

Midpoint = ((-7 + 5)/2, (-4 + 10)/2)
= (-1, 3).

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Question 14

L3 · SPR

Points A and B have coordinates (−5, 7) and (7, 2), respectively. What is the distance between A and B?

Answer: **13**

One number; use a fraction or decimal if needed.

Show your reasoning or calculator setup.

Horizontal change = 12; vertical change = -5.
Distance = sqrt(12^2 + (-5)^2)
= sqrt(169) = 13.

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Question 15

L3 · MC

The midpoint of segment AB is (1, −2). Point A has coordinates (−8, 6). What is the y-coordinate of point B?

- ○ **A** −8
- ○ **B** 2
- ◉ **C** −10
- ○ **D** 10

Show your reasoning or calculator setup.

(6 + y_B)/2 = -2
6 + y_B = -4, so y_B = -10.

☐ Flag for review

Question 16

L4 · MC

Segment AB has endpoints A(−2, −1) and B(6, 3). Which equation represents the perpendicular bisector of AB?

- ◉ **A** y = −2x + 5
- ○ **B** y = 2x − 3
- ○ **C** y = (1/2)x
- ○ **D** y = −2x + 1

Show your reasoning or calculator setup.

Midpoint = (2, 1); slope of AB = 4/8 = 1/2.
Perpendicular slope = -2.
y - 1 = -2(x - 2), so y = -2x + 5.

☐ Flag for review

D Mixed timed sprint

17–20 of 24

Set a 13-minute timer for Questions 17–24 together. Use about 10 minutes for a first pass, 2 minutes to revisit flags, and 1 minute to finish entering answers. Calculator allowed; no notes. Choose algebra or a graph deliberately. This is a pacing drill, not a Bluebook replica.

Question 17 MC

The line **y = 1.25x + 18** intersects the y-axis at (0, b). What is the value of b?

- ○ **A** 0
- ○ **B** 1.25
- ◉ **C** 18
- ○ **D** 19.25

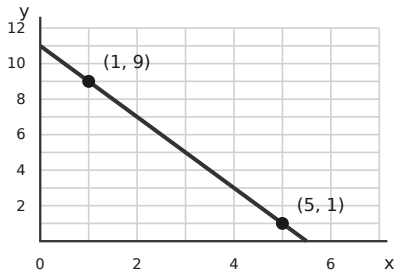
Show your reasoning or calculator setup.

At the y-axis, x = 0.
b = 1.25(0) + 18 = 18.

☐ Flag for review

Question 18 MC

The graph shows a line through the labeled points. What is the slope of the line?



- ○ **A** 2
- ○ **B** −1/2
- ○ **C** 1/2
- ◉ **D** −2

Show your reasoning or calculator setup.

m = (1 - 9)/(5 - 1) = -8/4 = -2.

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Question 19 MC

A delivery service models its total charge, in dollars, for n packages by **C(n) = 84 + 2.75n**. If the number of packages increases by 12, by how many dollars does the total charge increase?

- ◉ **A** 33
- ○ **B** 86.75
- ○ **C** 117
- ○ **D** 1,008

Show your reasoning or calculator setup.

Change in charge = 2.75(12) = 33 dollars.
The fixed $84 fee does not change.

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Question 20 SPR

The function h is defined by **h(x) = −2x + 9**. If h(x) = 17, what is x?

Answer **-4**

One number; use a fraction or decimal if needed.

Show your reasoning or calculator setup.

-2x + 9 = 17
-2x = 8, so x = -4.

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D Sprint, continued

21–24 of 24

Continue the same timer. Record a first-pass answer for every question before checking the key. For any calculator work, write enough of the setup that someone else can follow your method.

Question 21

MC

A line is perpendicular to **y = (3/5)x + 2** and passes through (−3, 4). Which equation represents the line?

- ○ **A** y = (−3/5)x + 11/5
- ◉ **B** y = (−5/3)x − 1
- ○ **C** y = (5/3)x + 9
- ○ **D** y = (−5/3)x + 4

Show your reasoning or calculator setup.

Perpendicular slope = -5/3.
y - 4 = (-5/3)(x + 3)
y = (-5/3)x - 1.

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Question 22

MC

Two studios charge for rehearsal time. Studio A charges a $45 booking fee plus $18 per hour. Studio B charges an $81 booking fee plus $12 per hour. For how many hours of rehearsal is the total charge the same at both studios?

- ○ **A** 2
- ○ **B** 3
- ○ **C** 4.5
- ◉ **D** 6

Show your reasoning or calculator setup.

45 + 18h = 81 + 12h
6h = 36, so h = 6 hours.

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Question 23

SPR

For a linear function f, **f(3) = 14** and **f(7) = 30**. If f(b) = b + 9, what is the value of b?

Answer: **7/3**

One number; use a fraction or decimal if needed.

Show your reasoning or calculator setup.

m = (30 - 14)/(7 - 3) = 4.
14 = 4(3) + c, so f(x) = 4x + 2.
4b + 2 = b + 9; 3b = 7; b = 7/3.

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Question 24

MC

The midpoint of the segment with endpoints (−4, 9) and (k, −3) lies on the line **y = −2x + 7**. What is the value of k?

- ○ **A** −8
- ○ **B** 4
- ◉ **C** 8
- ○ **D** 12

Show your reasoning or calculator setup.

Midpoint = ((k - 4)/2, 3).
3 = -2((k - 4)/2) + 7 = 11 - k.
k = 8.

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Repair the reasoning · Not scored

A student writes: “For $10x - 5y = 30$, dividing by -5 gives $y = -2x - 6$, so the slope is -2.” Identify the first incorrect step, write a correct equation, and explain why the slope sign makes sense.

First error → correct rewrite → sign check

First error: Dividing every term by -5 gives -2x + y = -6, not y = -2x - 6.

Correct rewrite: Add 2x to both sides to get y = 2x - 6. The slope is 2.

Sign check: If x increases by 1, 10x increases by 10; y must increase by 2

so that -5y decreases by 10 and 10x - 5y stays equal to 30.

Repair the model · After the timer

A plant’s height is modeled by $H(t) = 42 + 3.5t$, where H is centimeters and t is weeks. A student says its height after 4 weeks is 14 cm. What did the student actually calculate? Give the correct height and a one-sentence prevention rule.

Quantity calculated → requested quantity → prevention rule

The student calculated the increase in height: 3.5(4) = 14 cm.

Correct height: H(4) = 42 + 3.5(4) = 56 cm.

Prevention rule: For total height, add the starting height to the growth;

use rate times elapsed time alone only when asked for the increase.

Optional reference · Parts A–C only

$y = mx + b$

$m = (y_2 - y_1)/(x_2 - x_1)$

Parallel: equal slopes

Perpendicular: $m_1m_2 = -1$, when both slopes exist

Vertical □ horizontal

Midpoint: $((x_1 + x_2)/2, (y_1 + y_2)/2)$

Distance: $\sqrt{((\Delta x)^2 + (\Delta y)^2)}$

$f^{-1}(c)$ asks for the input x such that $f(x) = c$; it does not mean $1/f(c)$.

E Review, repair, transfer

10–15 min

Self-score after checking

One point per question. Repairs are unscored.

Part A / 6 Part B / 6 Part C / 4 Part D / 8 Total / 24

Log every missed or uncertain question; group questions with the same underlying error in one row. Suggested types: concept gap, translation, algebra/sign, response entry, or pacing. These totals are practice scores, not SAT scaled-score estimates.

Q number(s)	First incorrect step	Cause / type	Correct rule + next-time action

Reflection for the next session

My most important correction is...

For Q22, I chose algebra / a graph because...

The exact habit I will use next time is...

☐ All 24 answers entered ☐ Misses redone without the key ☐ Error log completed

Original instructor-created practice; not an official College Board assessment. Structure follows the supplied homework template; topics and visual palette follow the supplied Session 4 lesson. Format references: College Board SAT structure, Digital SAT specifications, and official SPR directions. No official questions reproduced.